human

Which buildings in Exeter have atleast 3 floors?

host_agent

Tool call: send_message - target: planning_agent - task_description: User wants to find buildings in Exeter with at least 3 floors. Provide the general steps to solve this geospatial query.

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings, and the condition is that these buildings must have at least 3 floors.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter as many search results", "Apply condition for buildings with at least 3 floors"]

host_agent

Tool call: send_message - target: named_area - task_description: Find the area polygon for Exeter.

named_area_agent

Tool call: call_os_ngd - filters: None - bbox: None - polygon_or_point: True - street_address: None - filename: exeter_area_polygon

Tool call result: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.., count of records fetched : 98.

Tool call: send_message - target: data_analysis_agent - task_description: Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'.

data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['exeter_area_polygon']

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where the name or description indicates 'Exeter' and is a settlement/city/town
    mask = (
        (gdf['name1_text'].str.contains('Exeter', case=False, na=False)) |
        (gdf['description'].str.contains('Exeter', case=False, na=False)) |
        (gdf['sameasdbpedia'].str.contains('Exeter', case=False, na=False)) |
        (gdf['sameasgeonames'].str.contains('exeter', case=False, na=False))
    ) & (
        (gdf['descriptiongroup'].str.contains('Settlement', case=False, na=False)) |
        (gdf['description'].str.contains('City|Town', case=False, na=False))
    )
    filtered = gdf[mask]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon specifically representing Exeter as a city or settlement.'
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered to 2 polygon(s) for Exeter. Artifact name: filtered_exeter_area_polygon Description: Polygon(s) from exeter_area_polygon specifically representing Exeter as a city or settlement.

named_area_agent

I found the area polygon for Exeter. The filtered artifact is named **filtered_exeter_area_polygon**, which specifically represents Exeter as a city or settlement.

host_agent

Tool call: send_message - target: buildings - task_description: Search for buildings with at least 3 floors within the area defined by the artifact 'filtered_exeter_area_polygon'.

buildings_agent

Tool call: call_os_ngd - filters: None - bbox: filtered_exeter_area_polygon - polygon_or_point: True - street_address: False - filename: exeter_buildings

Tool call result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for buildings with at least 3 floors within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", ...], counts of records fetched are : [69169, 69342, 85480].

Tool call: send_message - target: data_analysis_agent - task_description: Filter the artifact 'buildings_exeter_buildings' to find buildings with at least 3 floors. Save the filtered results as 'filtered_buildings_3_floors_exeter'.

data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['buildings_exeter_buildings']

Tool Code Executor

```
import geopandas as gpd

def filter_buildings_with_3_floors(data: list):
    gdf = data[0]
    # Filter for buildings with at least 3 floors (numberoffloors >= 3)
    filtered = gdf[gdf['numberoffloors'] >= 3]
    summary = f"Filtered buildings with at least 3 floors. Number of entries: {len(filtered)}."
    artifact_name = 'filtered_buildings_3_floors_exeter'
    artifact_description = 'GeoDataFrame of buildings in Exeter with at least 3 floors, including all original columns and geometry.'
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered buildings with at least 3 floors. Number of entries: 5,390. A new artifact has been generated with the name: filtered_buildings_3_floors_exeter. This artifact contains a GeoDataFrame of buildings in Exeter with at least 3 floors, including all original columns and geometry.

buildings_agent

I have filtered the buildings with at least 3 floors within the area defined by 'filtered_exeter_area_polygon'. The results are saved in the artifact named 'filtered_buildings_3_floors_exeter'.

host_agent

Tool call: generate_metadata_for_all_artifacts - args: {}

Tool call: send_message - target: plotting_agent - task_description: Plot the buildings from the artifact 'filtered_buildings_3_floors_exeter' to show all buildings in Exeter with at least 3 floors.

plotting_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['filtered_buildings_3_floors_exeter']

Tool Code Executor

```
def plot_buildings_3_floors_exeter(data: list):
    import geopandas as gpd
    import folium
    import pandas as pd
    import os

    # Get the first (and only) GeoDataFrame
    gdf = data[0].copy()

    # Convert all columns with Timestamp to string for folium compatibility
    for col in gdf.columns:
        if pd.api.types.is_datetime64_any_dtype(gdf[col]):
            gdf[col] = gdf[col].astype(str)

    # Ensure CRS is EPSG:4326 for folium
    if gdf.crs is not None and gdf.crs.to_string() != 'EPSG:4326':
        gdf = gdf.to_crs(epsg=4326)

    # Center map on Exeter (approximate center of all buildings)
    centroid = gdf.geometry.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')

    # Add all buildings to the map
    folium.GeoJson(
        gdf,
        name='Buildings (3+ floors)',
        tooltip=folium.GeoJsonTooltip(fields=['osid', 'description', 'numberoffloors', 'height_relativeroofbase_m'],
                                      aliases=['OSID', 'Type', 'Floors', 'Height (m)'],
                                      localize=True)
    ).add_to(m)

    # Save map to HTML
    map_filename = 'exeter_buildings_3_floors.html'
    m.save(map_filename)

    # Prepare summary (first 5 entries)
    summary_rows = gdf[['osid', 'description', 'numberoffloors', 'height_relativeroofbase_m']].head(5)
    summary = f"Plotted {len(gdf)} buildings in Exeter with at least 3 floors. First 5 entries:\n" + summary_rows.to_string(index=False)

    return [
        summary,
```

```
        'exeter_buildings_3_floors_map',
        'Folium map showing all buildings in Exeter with at least 3 floors, with tooltips for OSID, type, floors, and height.',
        map_filename
    ]
```

Plotted 5390 buildings in Exeter with at least 3 floors. First 5 entries: osid description numberoffloors height_relativeroofbase_m 2cd23260-72cb-4bd4-b05e-a97d95e8e10c Semi-Detached House 3.0 8.1 47b90969-3566-4013-a36b-70b33a749c25 Semi-Detached House 3.0 6.5 ea7b1764-bbac-48e7-8325-72523aa69799 Detached House 3.0 8.1 ca4245c7-8353-4e83-a0eb-c24bd1a98223 Multiple Residential Accommodation 4.0 5.1 7f5fd0e0-8166-45b6-9071-0ff1c53c7c97 Multiple Residential Accommodation 3.0 7.3

host_agent

Here are the results for your query:

- There are 5,390 buildings in Exeter with at least 3 floors.
- These buildings include various types such as semi-detached houses, detached houses, and multiple residential accommodations.
- The results include details like OSID, building type, number of floors, and relative roof base height.

A map has been generated showing all these buildings in Exeter, with tooltips for each building's details.

If you would like to view or download the map, let me know!